Apertus Court Decisions

Team beyondExp. Track 2B, Own Project / Apertus Adoption. Code and head weights Apache-2.0. This report CC-BY-4.0.

Summary

Apertus 1.5 8B stays frozen and reads the facts of a Swiss Federal Supreme Court ruling in German, French, or Italian. A sheaf head on the last hidden states returns one label from a closed list. Legal area is the usable task: 88.3% accuracy on 7577 official test cases, against 26.0% for always choosing Substantive Criminal. At confidence 0.90 or higher the head answers 72.5% of those cases and is right on 98.0% of that slice. Dismissal versus approval does not beat always saying dismissal (80.92% against 80.93%). The system is a triage aid for the legal area when it is confident. It is not a case decider and it is not legal advice.

Architecture

The reader is swiss-ai/Apertus-v1.5-8B. Its weights are not updated. The trained module is a sheaf glue head: two maps of shape 4096 by 4096, both initialised as identity, a positive scale, and a bias on the pooled answer. The score of an answer is the maximum cosine between any document token and any token of that answer, plus the bias. The choice is a softmax over the closed answer list.

Target architecture

The runtime is on-premise. Scoring needs the frozen 8B weights and this head on local disk, and it does not call a network. The same layout is the air-gapped layout once the weights are inside the boundary. The same image can run on a sovereign Swiss cloud host, with the weights mounted there rather than fetched per request. A chat endpoint cannot drive the head, because the head scores last-layer hidden states.

Use of Apertus

Apertus is the encoder. One training pass used 63328 original-language rulings, seed 1, AdamW learning rate 0.001, token cap 2048. Mean training loss was 0.527 on outcomes and 0.285 on legal area. The published head is float16. make run loads it and recomputes the tables below from the saved predictions.

Data

Sources are rcds/swiss_judgment_prediction and rcds/swiss_law_area_prediction, German, French, and Italian, CC BY-SA 4.0. The input is the facts section. The label is the court's own. The reasoning section is left out. The English machine-translated split is not used. Train counts are 53474 judgments and 9854 law-area rulings. This repository does not redistribute the rulings.

Evaluation

Official test, legal area: accuracy 88.31%, macro-F1 0.638, majority accuracy 26.04%, majority macro-F1 0.032, mean loss 0.421.

label	support	predicted	P	R	F1
Bankruptcy	206	185	0.951	0.854	0.900
Competition and Antitrust	6	0	0.000	0.000	0.000
Criminal Procedure	1011	1319	0.732	0.955	0.828
Employment Contract	389	402	0.958	0.990	0.973
Expropriation	0	0	0.000	0.000	0.000
Family	263	160	1.000	0.608	0.757
Intellectual Property	11	7	1.000	0.636	0.778
Other Fiscal	1	0	0.000	0.000	0.000
Public Administration	280	354	0.305	0.386	0.341
Rental and Lease	846	874	0.960	0.992	0.976
Substantive Criminal	1973	1903	0.975	0.941	0.958
Tax	1580	1309	0.938	0.777	0.850
Urban Planning and Environmental	1011	1064	0.909	0.956	0.932

Official test, outcome: accuracy 80.92%, macro-F1 0.462, majority accuracy 80.93%, majority macro-F1 0.447, mean loss 0.412.

label	support	predicted	P	R	F1
dismissal	11786	14475	0.811	0.996	0.894
approval	2777	88	0.489	0.015	0.030

At confidence 0.90 the legal-area head covers 72.5% of test cases (n 5493) at accuracy 98.0%. The outcome head covers 34.5% (n 5027) and every prediction in that slice is dismissal, including 135 cases the court approved.

Largest legal-area confusions: Tax read as Criminal Procedure (136); Substantive Criminal read as Criminal Procedure (107); Tax read as Public Administration (99); Public Administration read as Criminal Procedure (89); Family read as Public Administration (87).

Test accuracy by language. Macro-F1 is not a ranking across languages, because the label mix differs.

task	lang	n	accuracy	macro-F1	majority
legal area	de	2248	87.6%	0.399	42.5%
legal area	fr	4579	87.7%	0.511	40.4%
legal area	it	750	94.1%	0.221	77.6%
outcome	de	8270	80.6%	0.466	80.5%
outcome	fr	5569	81.1%	0.457	81.3%
outcome	it	724	82.7%	0.453	82.7%

Official validation: legal area 93.42% on 2872 cases against majority 30.61%, macro-F1 0.685. Outcome 79.86% on 7041 cases against majority 79.83%.

Limitations

The outcome head must not be shown as a decision. Legal area is weak on public administration, and some tax cases are read as criminal procedure. Competition and antitrust, and other fiscal, have too little support to learn. Another epoch on this set will not create those rare classes. The head is not a substitute for a lawyer, a clerk, or the court.

Reproducibility

From track_2b, make run builds the image, loads checkpoint/head.pt, and checks that data/predictions.jsonl matches data/metrics.json. A new ruling needs a local forward of frozen Apertus 1.5 8B, then SheafGlueHead.glue on those hidden states and on the embeddings of the allowed answers. Training was one epoch, seed 1, learning rate 0.001, token cap 2048.

Next steps

Show a legal-area label only at confidence 0.90 or higher. Send the rest, and every outcome question, to a person. Review the public-administration and criminal-procedure boundary before relying on those two names.